

CityLearn 2022 Project - Building 2 Energy Analysis

Objectives:

1. To analyze the electricity consumption of Building 2 using the provided dataset.
2. To calculate the solar energy generated by the PV system and convert it to kWh.
3. To determine the total electricity bill without considering PV generation.
4. To calculate the net electricity consumption after considering solar generation.
5. To identify the surplus energy (wasted or exportable) and understand energy trends in the building.
6. To visualize and interpret the results using basic Python constructs like lists and loops.

Step 1: Load the CSV data manually

- We open the CSV files using Python's open function.
- The first line is skipped because it contains the header.
- Each line is split at the comma and stored in a list.
- This gives us all the building and price data to work with.
- We now have two lists: one for building data and one for prices.

```
In [3]: # reading CSV manually into lists

building_data = []
with open(r"building_2.csv") as file:
    next(file) # skip header
    for line in file:
        values = line.strip().split(",")
        building_data.append(values)
```

```
In [4]: price_data = []
with open("price.csv", "r") as f:
    next(f) # skip header
    for line in f:
        price_data.append(float(line.strip()))
```

Step 2: Extract non-shiftable load and PV generation

- We separate the building consumption and solar generation values.
- The first column is saved as non-shiftable load.
- The second column is saved as PV generation per kW.
- We convert strings to floats so we can do calculations.
- Now both lists are ready for further processing.

```
In [5]: # Extract Equipment Electric Power [kWh] as non-shiftable Load

non_shiftable_load = []
for row in building_data:

    # assuming first column is Equipment Power
    non_shiftable_load.append(float(row[0]))
```

```
In [6]: # Extract Solar Generation [W/kW]

pv_per_kw = []
for row in building_data:

    # assuming second column is Solar Generation
    pv_per_kw.append(float(row[1]))
```

Step 3: Convert PV generation to kWh

- The PV data is in W/kW, so we need to convert it to kWh.
- We multiply by the installed capacity and divide by 1000.
- This gives the actual solar energy produced each hour.
- The values are stored in a new list called net solar generation.
- Now we can compare it directly with the building's consumption.

```
In [7]: installed_kw = 1
net_solar_generation = []
for pv in pv_per_kw:
    net_solar_generation.append(pv * installed_kw / 1000)
```

Step 4: Total consumption and bill without PV

- We sum the non-shiftable load to get total electricity used.
- Then we multiply each hour's consumption by the price to get the bill.
- All hourly bills are stored in a list called bill without PV.
- The sum of this list gives the total electricity cost.
- This shows how much the building would pay without solar.

```
In [8]: # Total electricity consumption

net_consumption_total = 0
for load in non_shiftable_load:
    net_consumption_total += load

print("Building net electricity consumption (kWh):", net_consumption_total)
```

Building net electricity consumption (kWh): 54.199999999999996

```
In [9]: # Calculate bill without PV

bill_without_pv = []
for i in range(len(non_shiftable_load)):
    bill_without_pv.append(non_shiftable_load[i] * price_data[i])
```

```
In [10]: # Calculate Total bill without PV

total_bill_without_pv = 0
for bill in bill_without_pv:
    total_bill_without_pv += bill
```

```
In [11]: print("Total bill without PV ($):", total_bill_without_pv)
```

Total bill without PV (\$): 7.5840000000000005

Step 5: Net consumption after PV

- We subtract the solar generation from the building load each hour.
- Positive values mean electricity is still needed from the grid.
- Negative values mean extra energy produced (wasted or stored).
- We separate them into net electricity with PV and wasted energy lists.

- Finally, we calculate the sum of both lists for totals.

```
In [12]: net_consumption = []
        for i in range(len(non_shiftable_load)):
            net_consumption.append(non_shiftable_load[i] - net_solar_generation[i])
```

```
In [13]: net_electricity_with_pv = []
        wasted_energy = []
```

```
In [14]: for val in net_consumption:
        if val >= 0:
            net_electricity_with_pv.append(val)
            wasted_energy.append(0)
        else:
            net_electricity_with_pv.append(0)
            wasted_energy.append(val)
```

```
In [15]: # Calculate sums

        net_electricity_sum = 0
        for val in net_electricity_with_pv:
            net_electricity_sum += val
```

```
In [16]: wasted_energy_sum = 0
        for val in wasted_energy:
            wasted_energy_sum += val
```

```
In [17]: print("Net electricity consumption with PV (kWh):", net_electricity_sum)
        print("Wasted energy (kWh):", wasted_energy_sum)
```

Net electricity consumption with PV (kWh): 44.74999999999999

Wasted energy (kWh): 0

Step 6: Simple text-based visualization

- We print the first 10 values of each list to see trends.
- This shows building load, solar generation, net consumption, and wasted energy.
- Even without graphs, we can understand the energy profile.
- It helps to quickly check if the calculations make sense.
- This is useful for interpreting the results step by step.

```
In [18]: print("\nFirst 10 entries of non-shiftable load:", non_shiftable_load[:10])
```

First 10 entries of non-shiftable load: [5.2, 4.8, 5.5, 6.0, 5.0, 4.7, 5.1, 5.8, 6.2, 5.9]

```
In [19]: print("First 10 entries of net solar generation:", net_solar_generation[:10])
```

First 10 entries of net solar generation: [0.8, 0.9, 0.95, 1.0, 0.85, 0.7, 0.95, 1.1, 1.2, 1.0]

```
In [20]: print("First 10 entries of net consumption:", net_consumption[:10])
```

First 10 entries of net consumption: [4.4, 3.9, 4.55, 5.0, 4.15, 4.0, 4.1499999999999995, 4.699999999999999, 5.0, 4.9]

Conclusion

- Building 2's electricity consumption was analyzed using the dataset.
- PV generation was converted to kWh and compared with building load.
- Solar energy reduces the net electricity drawn from the grid.
- The total electricity bill without PV was calculated for comparison.
- Some energy is wasted when solar generation exceeds building demand.
- Using basic Python lists and loops, all calculations were done step by step.
- The study helps understand energy trends and the effect of PV on building electricity usage.